- Jetico BestCrypt Volume Encryption (*www.jetico.com/products/personal-privacy/bestcrypt-volume-encryption*) provides WDE for older MS-DOS and current Windows systems.

With improved encryption methods, extracting digital evidence will become more difficult. Because of these challenges, you need to know how to make remote live acquisitions, discussed in Chapter 10.

Understanding the Windows Registry

When Microsoft created Windows 95, it consolidated initialization (`.ini`) files into the **Registry**, a database that stores hardware and software configuration information, network connections, user preferences (including usernames and passwords), and setup information. The Registry has been updated and is still used in Windows Vista and later.

For investigative purposes, the Registry can contain valuable evidence. To view the Registry, you can use the Regedit (Registry Editor) program for Windows 9x and Regedt32 for Windows 2000, XP, and Vista. For Windows 7 and 8, both Regedit and Regedt32 are available.

Tip

For more information on using Regedit and Regedt32, see the Windows Resource Kit documentation for the OS version. You can find it at *https://support.microsoft.com/en-us/help/141377/differences-between-regedit.exe-and-regedt32.exe*.

In general, you can use the Edit, Find menu command in Registry Editor to locate entries that might contain trace evidence, such as information identifying the last person who logged on to the computer, which is usually stored in user account information. You can also use the Registry to determine the most recently accessed files and peripheral devices. In addition, all installed programs store information in the Registry, such as Web sites accessed, recent files, and even chat rooms accessed.

As a digital forensics investigator, you should explore the Registry of all Windows systems. On a live system, be careful not to alter any Registry setting to avoid corrupting the system and possibly making it unbootable.

Note

Several third-party tools, such as FTK Registry Viewer, are also available for accessing the Registry.

Exploring the Organization of the Windows Registry

The Windows Registry is organized in a specific way that has changed slightly with each new version of Windows. However, the major Registry sections have been consistent, with some minor changes, since Windows 2000; they're slightly different in Windows 9x/Me. Before proceeding, review the following list of Registry terminology:

- *Registry*—A hierarchical database containing system and user information.
- *Registry Editor*—A Windows utility for viewing and modifying data in the Registry. There are two Registry Editors: Regedit and Regedt32 (introduced in Windows 2000).
- *HKEY*—Windows splits the Registry into categories with the prefix HKEY_. Windows 9x systems have six HKEY categories and Windows 2000 and later have five. Windows programmers refer to the "H" as the handle for the key.
- *Key*—Each HKEY contains folders referred to as keys. Keys can contain other key folders or values.
- *Subkey*—A key displayed under another key is a subkey, similar to a subfolder in Windows or File Explorer.
- *Branch*—A key and its contents, including subkeys, make up a branch in the Registry.
- *Value*—A name and value in a key; it's similar to a file and its data content.
- *Default value*—All keys have a default value that may or may not contain data.
- *Hives*—Hives are specific branches in HKEY_USER and HKEY_LOCAL_MACHINE. Hive branches in HKEY_LOCAL_MACHINE\Software are SAM, Security, Components, and System. For HKEY_USER, each user account has its own hive link to `Ntuser.dat`.

The next piece of the puzzle is learning where data files that the Registry reads are located. The number of files the Registry uses depends on the Windows version. In Windows 9x/Me, it uses only two files, `User.dat` and `System.dat`. In Windows NT and later, there are six files: `Ntuser.dat`, `System.dat`, `SAM.dat`, `Software.dat`, `Security.dat`, and `Default.dat`. When examining Registry data from a suspect drive after you have made an acquisition and are reviewing it in a forensics tool, you need to know the location of these files. Table 5-6 shows how Registry data files are organized and explains these files' purposes in Windows Vista and later. For information on older Windows Registry files, see *http://support.microsoft.com/kb/250410*.

Table 5-6 Registry file locations and purposes

Filename and location	Purpose of file
Users*user-account*\\`Ntuser.dat`	User-protected storage area; contains the list of most recently used files and desktop configuration settings
Windows\\system32\\config\\`Default.dat`	Contains the computer's system settings
Windows\\system32\\config\\`SAM.dat`	Contains user account management and security settings
Windows\\system32\\config\\`Security.dat`	Contains the computer's security settings
Windows\\system32\\config\\`Software.dat`	Contains installed programs' settings and associated usernames and passwords
Windows\\system32\\config\\`System.dat`	Contains additional computer system settings
Windows\\system32\\config\\systemprofile	Contains additional NTUSER information

When viewing the Registry with Registry Editor, you can see the HKEYs used in Windows (see Figure 5-26).

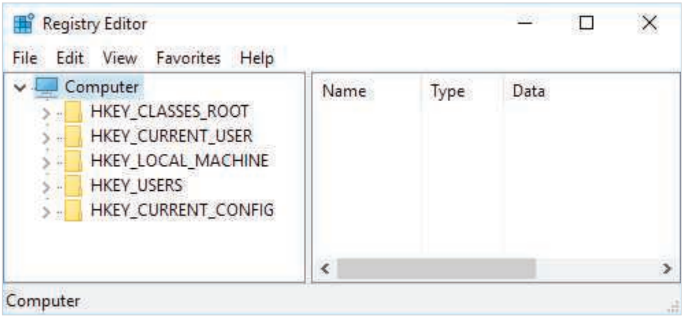

Figure 5-26 Viewing HKEYs in Registry Editor

Table 5-7 describes the functions of Registry HKEYs.

Table 5-7 Registry HKEYs and their functions

HKEY	Function
HKEY_CLASSES_ROOT	A symbolic link to HKEY_LOCAL_MACHINE\SOFTWARE\Classes; provides file type and file extension information, URL protocol prefixes, and so forth
HKEY_CURRENT_USER	A symbolic link to HKEY_USERS; stores settings for the currently logged-on user
HKEY_LOCAL_MACHINE	Contains information about installed hardware and software
HKEY_USERS	Stores information for the currently logged-on user; only one key in this HKEY is linked to HKEY_CURRENT_USER

(*continues*)

Table 5-7 Registry HKEYs and their functions (*continued*)

HKEY	Function
HKEY_CURRENT_CONFIG	A symbolic link to HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Hardware ProfileVxxxx (with xxxx representing the current hardware profile); contains hardware configuration settings
HKEY_DYN_DATA	Used only in Windows 9x/Me systems; stores hardware configuration settings

Examining the Windows Registry

Some forensics tools, such as X-Ways Forensics, OSForensics, Forensic Explorer, and FTK, have built-in or add-on Registry viewers. For this next activity, your company's Legal Department has asked you to search for any references to any e-mail addresses containing the name Denise or Robinson with the domain name outlook.com. A paralegal gives you a raw (dd) image file containing `InCh05.img`, a forensic image of a Windows 8 computer's hard drive used by Superior Bicycle employee Denise Robinson.

For this activity, you use OSForensics to examine Denise Robinson's `NTUser.dat` file. If you find any items of interest, add them to an OSForensics case report that you can give to the paralegal. The following steps explain how to generate a case report in OSForensics.

Note

Before beginning this activity, download the zipped file in the downloads section for this chapter on the student companion site to your *Work*\Chap05\Chapter folder. When the download is finished, double-click the zipped file in File Explorer to extract the `InChap05.img` file. The work folder pathname you see in screenshots might differ.

To examine Registry files with OSForensics, follow these steps:

1. Start OSForensics with the **Run as administrator** option, and click **Continue Using Trial Version**.
2. In the left pane, click **Manage Case**, if necessary. In the Manage Case pane on the right, click the **New Case** button. In the New Case dialog box, type **InChap05** in the Case Name text box and your name in the Investigator text box. For the Acquisition Type setting, click the **Investigate Disk(s) from Another Machine** option button (see Figure 5-27). Click **Custom Location** for the Case Folder option. Click the **Browse** button on the lower right, navigate to and click your work folder, and then click **OK** twice.

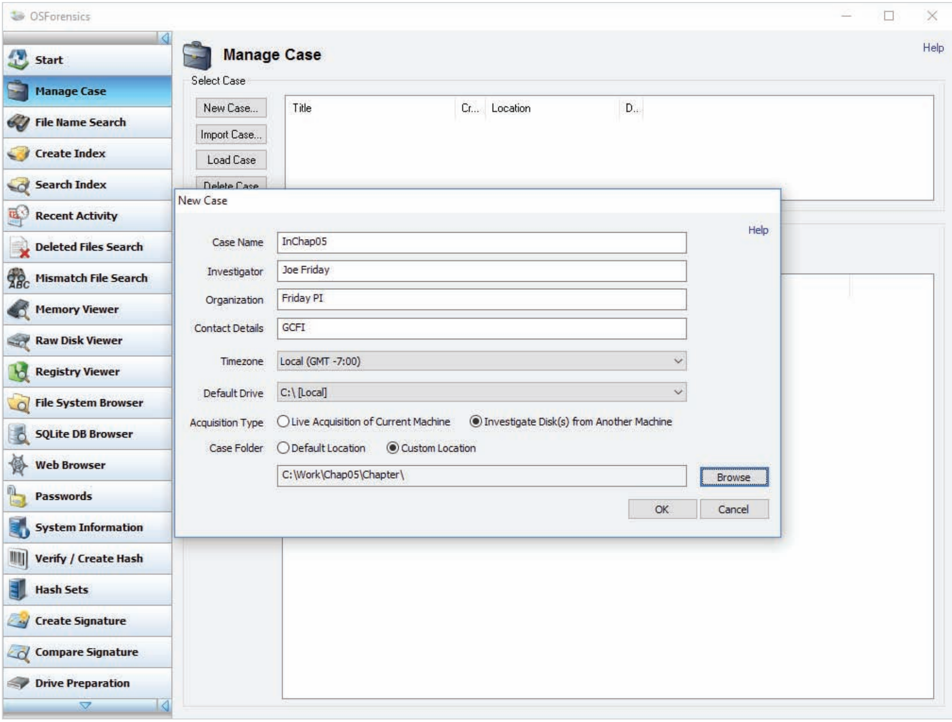

Figure 5-27 The New Case dialog box

Source: PassMark Software, *www.osforensics.com*

> **Tip** ⓘ
>
> Notice the drive letter in the OSFMount - Mount drive dialog box, which you use in Step 3. This image is mounted as read-only as an attached drive on your computer and becomes accessible to OSForensics.

3. Copy the **`InChap05.img`** file to the ***Work*\Chap05\Chapter** folder, if necessary. To mount this disk image, scroll down the navigation bar on the left, and click **Mount Drive Image**. In the Mounted virtual disks window, click the **Mount new** button. In the OSFMount - Mount drive dialog box that opens (see Figure 5-28), click the ... button next to the Image file text box, navigate to your work folder, click **`InCh05.img`**, click **Open**, and then click **OK**.
4. In the navigation bar on the left, click **Registry Viewer**. In the "Select registry hive file to open" dialog box, click the **Select Drive** list arrow, and then click the drive letter that was shown in Step 3 (see Figure 5-29). The drive letter on your

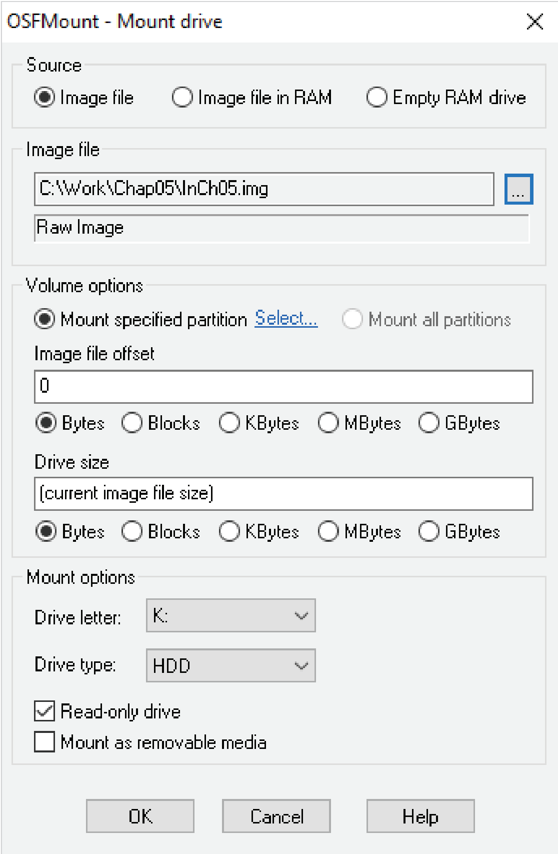

Figure 5-28 Mounting a drive in OSForensics

Source: PassMark Software, *www.osforensics.com*

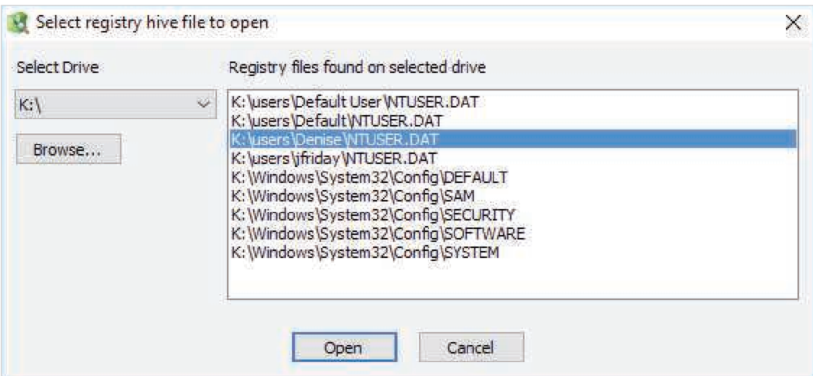

Figure 5-29 The "Select registry hive file to open" dialog box

Source: PassMark Software, *www.osforensics.com* system is likely to be different. In the list of files on the right, click ***DriveLetter\users\Denise\NTUSER.DAT***, and then click **Open**.

5. In the OSForensics Registry Viewer, click **Search**, **Find** from the menu to open the Find dialog box. In the Search For text box, type **Outlook.com** (see Figure 5-30), and then click the **Find** button.

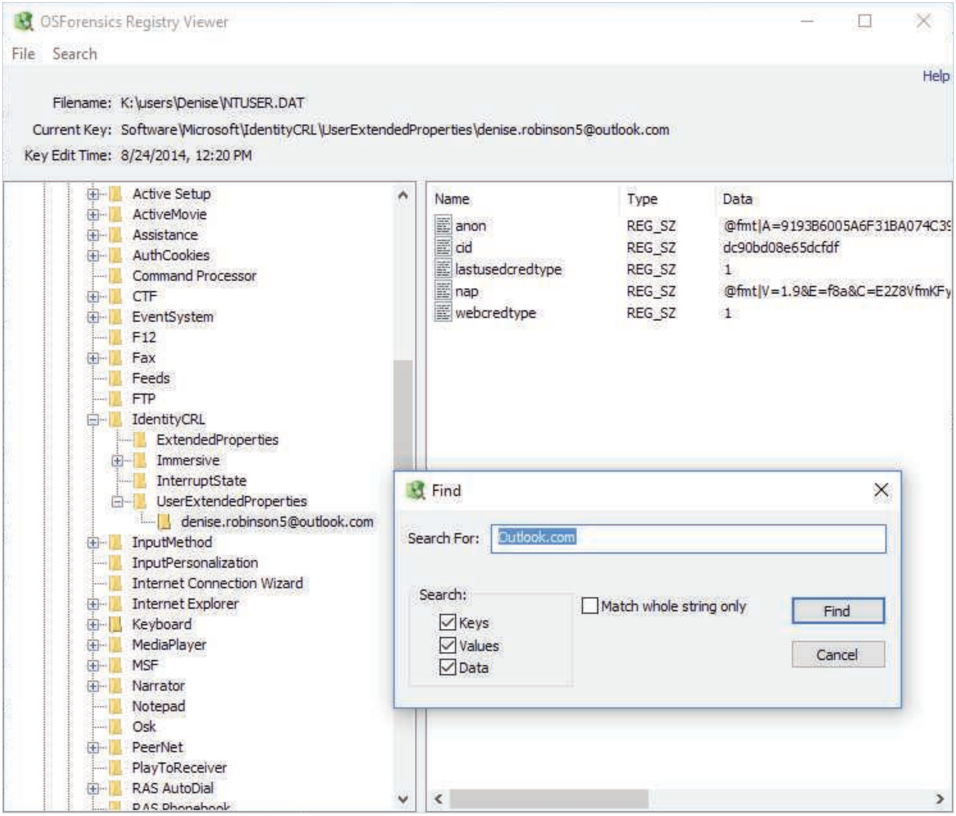

Figure 5-30 Entering a search term

Source: PassMark Software, *www.osforensics.com*

6. In the Registry Viewer pane on the right, right-click the first search hit and click **Add to Case** to open the Please Enter New Case Item Details dialog box. In the Title text box, type **Outlook e-mail address for Denise Robinson**, and then click **OK**.

7. In the Find dialog box, click **Find** again. Right-click the next search hit and click **Add to Case**. Type **Outlook e-mail Web site** in the Title text box, and then click **OK**.

8. In the Find dialog box, click **Find** again. Right-click the next search hit and click **Add to Case**. Type **Denise Robinson's e-mail address** in the Title text box, and then click **OK**. Exit Registry Viewer.

9. In the main OSForensics window, click **Manage Case** in the navigation bar on the left. In the Manage Current Case pane on the right, click the **Generate Report** button. In the Export Report window, click **Browse** next to the Output Location text box, navigate to your work folder, and click **OK** to open the report in your Web browser. Scroll to the bottom of the navigation bar and click **Exit**. In the "Mounted virtual disks" window, click the **Dismount all & Exit** button. Click **Yes** in the OSMount warning message, and then click **OK** again to exit OSMount.
10. Review the contents of your report, and then exit your Web browser.

An extensive amount of information is stored in the Registry. With Registry data, you can ascertain when users went online, when they accessed a printer, and many other events. A lot of the information in the Registry is beyond the scope of this book, so you're encouraged to expand your knowledge by attending training sessions or classes.

Understanding Microsoft Startup Tasks

You should have a good understanding of what happens to disk data at startup. In some investigations, you must preserve data on the disk exactly as the suspect last used it. Any access to a computer system after it was used for illicit purposes alters your disk evidence. As you learned in Chapter 3, altering disk data lessens its evidentiary quality considerably. In some instances, accessing a suspect computer incorrectly could make the digital evidence corrupt and less credible for litigation.

In the following sections, you learn what files are accessed when Windows starts. This information helps you determine when a suspect's computer was last accessed, which is particularly important with computers that might have been used after an incident was reported.

Startup in Windows 7, Windows 8, and Windows 10

Since Windows Vista, Microsoft has changed its approach to OS boot processes. In addition, Windows 8 and 10 are multiplatform OSs that can run on desktops, laptops, tablets, and smartphones. This discussion covers desktop and laptop computers running Windows 10, although Windows Vista, 7, and 8 are very similar.

All Windows 8 and 10 boot processes are designed to run on multiple devices, ranging from desktop or laptop systems to tablets and smartphones. In Windows Vista and later, the boot process uses a boot configuration data (BCD) store. For desktops and laptops (BIOS-designed systems), a BCD Registry file in the \Boot\Bcd folder is maintained to control the boot process. To access this file, you use the BCD Editor; Regedit and Regedt32 aren't associated with this file.

In Windows 8 and 10, the BCD contains the boot loader that initiates the system's bootstrap process when Windows starts. To access the Advanced Boot Options menu